



Negotiating creator identity: agency and ethical awareness in AI-assisted art education

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Abstract

Artificial intelligence (AI) is influencing creative work, negotiating not only how artists create but also how they understand their own identity as creators. This study examines how art high school students navigate AI's role in their creative processes, using identity formation theory as a framework. While existing research primarily focuses on AI's technical benefits, such as efficiency and productivity, there has been little exploration of how AI alters young creators' self-concept and autonomy. This study investigates whether students perceive AI as a mere tool, a collaborator, or a potential threat to their creative identity. It also explores how AI's increasing presence affects students' confidence in their originality, their sense of authorship, and their ethical concerns about AI-generated content, particularly regarding copyright and creative ownership. The findings reveal that students have mixed views on AI: some embrace it as a source of inspiration and efficiency, while others express discomfort about its potential to undermine human creativity. These results highlight the need for AI education that extends beyond technical training to include discussions on ethics, authorship, and career adaptability in an AI-driven creative industry. By understanding how students construct their creative identities in relation to AI, educators and policymakers can better support them in navigating an evolving digital landscape.

Keywords artificial intelligence · art education · creator identity · identity formation · ethical awareness

1 Introduction

The rapid adoption of artificial intelligence (AI) has transformed a wide range of domains, including education and the creative industries. In creative contexts, AI presents a qualitatively different challenge from traditional digital technologies. Rather than simply supporting task performance, generative AI actively participates in idea generation and

content creation, thereby influencing the nature of creative work itself (Millet et al. 2023; Shin 2021).

This shift complicates the process of technology adoption. As AI becomes embedded in creative workflows, users may no longer perceive it as a passive tool, but as an active agent that influences both process and outcome (Jiang et al. 2024; Lee and Park, 2024). In artistic domains, this transformation raises fundamental questions about authorship, originality, and the role of human (Shin 2026b). The growing presence of generative AI in creative industries has intensified tensions between automation and artistic authorship. While AI offers practical benefits such as efficiency, speed, and inspiration, it also introduces concerns regarding ownership, fairness, and the legitimacy of AI-assisted work. These tensions are particularly salient for adolescents, who are in the process of forming their identities as creators and future professionals.

Despite the rapid integration of AI into creative practices, existing research has largely focused on technical skills, tool use, and efficiency-oriented outcomes. Relatively little attention has been paid to how young learners interpret AI's role in creative processes, or how these experiences shape their

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creator identity, sense of agency, and ethical reasoning (Lee et al., 2025). This gap is especially critical in art education, where identity formation is a central component of learning.

To address these gaps, the present study examines how high school art students experience and interpret AI integration in a classroom setting. Specifically, this study investigates the following research questions:

RQ1. How do art high school students' perceptions of AI's impact on art, careers, and creative work change after participating in an AI-supported class?

RQ2. How do students conceptualize AI's role in the creative process, as a tool, a collaborator, or a threat to creator identity?

RQ3. How does AI integration negotiate students' perceptions of creator identity and agency?

RQ4. How do students' ethical concerns about AI interact with their acceptance of AI and their self-perception as future creators?

These questions foreground the tensions students experience between the practical affordances of AI (e.g., efficiency, speed, and creative support) and their ethical concerns and identity-related uncertainties.

By addressing these questions, this study provides insight into how young creators make sense of AI in artistic contexts. The findings contribute to research on AI education by moving beyond skill acquisition to examine the interplay among competence, identity, and ethics, and offer implications for designing learning environments that support students in navigating human-AI collaboration.

2 Literature review

2.1 AI acceptance of AI technologies

When new technologies emerge and develop, user acceptance of those technologies has become an important topic. One of the most widely used theoretical frameworks in these discussions is the Technology Acceptance Model (TAM) (Davis 1989; Choung et al. 2023; Kelly et al. 2023). The TAM explains that technology acceptance is primarily influenced by perceived usefulness and perceived ease of use (Choung et al. 2023; Shin 2021). In other words, the more users perceive a technology as improving efficiency and productivity, the higher the likelihood they will accept it (Gerlich 2023; Sohn and Kwon 2020). Trust in technology is also discussed as a crucial factor influencing the acceptance of AI (Mcknight et al. 2011). This trust is formed based on perceptions regarding the reliability, transparency, and fairness of the system (Rane et al. 2024). These technology acceptance factors also apply to AI, particularly in how users understand and trust the decision-making processes and outcomes of AI

systems (Rane et al. 2024). Based on this perspective, existing studies have primarily relied on the TAM framework to explain user acceptance of technology (Jiang et al. 2024; Kelly et al. 2023). In particular, many studies related to AI acceptance tend to focus on how to facilitate user acceptance (Rane et al. 2024). This approach presupposes AI as a functional tool, and discussion on how users understand and interpret the role of AI is relatively limited. These studies also do not always present consistent results regarding AI acceptance (Jiang et al. 2024).

AI is developing at a much faster pace than existing technologies and is demonstrating a new technological paradigm that actively intervenes in task performance, moving beyond technology used merely passively. These changes increase the likelihood that AI will be perceived not merely as an execution tool, but as an entity that intervenes in human judgment and creative processes (Millet et al. 2023). Due to these characteristics, it is argued that understanding AI acceptance requires discussion within a broader context that extends beyond TAM (Jiang et al. 2024; Kelly et al. 2023). For example, while some studies report high levels of user acceptance of AI technology, others find significant levels of resistance (Jiang et al. 2024). These conflicting results may be due to the fact that attitudes toward AI can vary depending on how users perceive the role of AI and what the context is.

Jiang et al. (2024) suggest that attitudes toward AI acceptance may differ depending on whether users perceive AI as a service provider that assists human activities or as a task replacement system. As such, the perception of the role of AI goes beyond mere acceptance and is also connected to how users define their own roles. In fact, AI systems that support human activities, such as shopping recommendations or customer service, and AI systems that directly replace human labor can elicit different user reactions. Furthermore, studies have reported that user backlash may occur if AI is perceived as a threat to uniquely human cognitive abilities, such as artistic creativity (Longoni et al. 2019; Millet et al. 2023). This reaction is not merely a rejection of technology but is related to the perception that unique human creativity or expertise is under threat, and it can be particularly strong in areas closely related to identity, such as creative work. (Millet et al. 2023; Strich et al. 2021). These reactions may be related to perceptions of threats to professional identity or technological overload, and these factors can act as deterrents to AI-based systems (Kim et al. 2022; Strich et al. 2021).

Overall, user attitudes and perceptions regarding AI acceptance may be influenced by various contextual factors rather than exhibiting consistent patterns (Jiang et al. 2024). AI acceptance, therefore, needs to be understood not merely as the usefulness or ease of use of the technology, but as a process of interpreting how users perceive AI and how it

relates to their own roles (Jiang et al. 2024). Rather than being explained solely by factors such as technical characteristics, perceived utility, or ease of use, AI acceptance should be examined from a more multifaceted perspective, including how users interpret the role and significance of AI systems. Particularly given the increasing use of AI in K-12 classrooms, a deeper understanding and discussion regarding learners' AI acceptance is necessary.

2.2 Human–AI agency in learning environments

Human agency refers to an individual's ability to act independently and influence their environment (Bandura 2001; Mouta et al. 2025). Humans are described as autonomous beings who shape their own behavior through abilities such as intentionality, self-regulation, prediction, and self-reflection (Bandura 2006). This agency is understood not merely as a characteristic of the individual, but as a process formed within social relationships and temporal contexts (Bandura 2006). This perspective is also significantly applicable in the context of education. In other words, learners can be understood not merely as recipients of knowledge, but as agents who actively participate in their own learning process.

Beyond questions of acceptance, the growing involvement of AI in decision-making and content generation raises important questions about agency. This is particularly significant because agency has traditionally been understood as a uniquely human capacity (Schreiber, 2024). As AI has moved beyond being a passive tool and begun to actively intervene in human activities through content creation, decision support, and problem-solving, this assumption is increasingly challenged (Krakowski 2025). Due to these changes, some studies understand AI technology from the perspective of human-AI interaction (e.g., Courtois and Timmermans 2018) or propose the concept of machine agency (e.g., Pickering et al. 2017; Sundar 2020). In particular, since human and machine agencies coexist in AI-based environments, tensions regarding the distribution of authority and control may arise between humans and AI (Jia et al. 2012; Kang and Lou 2022; Kang and Kim 2020). For example, while users may positively evaluate the efficiency and convenience provided by AI, they may simultaneously experience concerns that human control may be weakened (Jia et al. 2012; Kang and Kim 2020).

This tension extends beyond simple functional interaction between humans and AI to the question of 'who makes the decision' and 'who takes responsibility' regarding judgments when using AI (Jia et al. 2012; Kang and Kim 2020). In other words, learners may perceive AI as a simple tool or as an entity that influences decision-making, and this difference in perception directly affects the agency experience. Furthermore, human-AI interaction can manifest in various forms, such as cooperation, control, and resistance. Representative

concepts explaining these relationships are automation and augmentation (Krakowski 2025; Raisch and Krakowski 2021). Automation refers to AI replacing tasks previously performed by humans, while augmentation refers to humans and AI collaborating to expand upon those tasks (Raisch and Krakowski 2021). This perspective suggests the potential for AI technology to evolve into a hybrid agency where humans and AI work together to solve problems, rather than completely replacing humans (Raisch and Krakowski 2021). This distinction between automation and augmentation is closely linked to how learners perceive AI and define their own roles (Raisch and Krakowski 2021). For instance, if AI is perceived as an auxiliary tool, agency is maintained; however, if AI is perceived as driving the outcome, it can lead to a loss of control (Jia et al. 2012; Kang and Kim 2020).

This discussion holds significant implications in the context of education. This is because, as AI technology actively intervenes in learning support, content creation, and problem-solving processes, the learner's role and mode of control may also change. Consequently, in educational research, it is becoming increasingly important to understand learners as active agents and to adopt a perspective that considers their agency (Mouta et al. 2025). This perspective is particularly important for adolescent learners, as they are in a critical period for identity formation where individuals explore their abilities and roles (Erikson 1988). Adolescent learners form diverse identity experiences while exploring their capabilities and potential through new technologies and creative activities (Kay 2018). However, there has not yet been sufficient discussion on how learners experience and interpret their agency in an AI-based environment.

Furthermore, AI technology exhibits characteristics distinct from existing technologies. For instance, unlike traditional technologies where users select and execute specific functions (e.g., coloring), generative AI demonstrates a top-down interaction style that automatically generates results beyond user input (Mouta et al. 2025). This top-down interaction alters the structure of interaction between users and technology, raising concerns that it may weaken learners' agency. Changes in this interaction structure can have a direct impact on the sense of control and agency that learners experience in their creative process (Mouta et al. 2025). Therefore, in the context of AI education, an approach that considers learners' agency and participation experiences alongside the functionality of the technology itself is necessary (Mouta et al. 2025).

2.3 Creator identity formation

Identity formation is described as a developmental process in which an individual seeks to understand who they are and how they relate to society (Erikson 1988; Hamman and Hendricks 2005). Erikson's psychosocial development

theory explains human development as a lifelong process of identity formation (Erikson 1968; Hamman and Hendricks 2005). Individuals form their values and roles through various social experiences and role exploration. This process of identity formation takes place not only through internal exploration but also through interaction with social and cultural contexts.

Particularly, adolescence is understood as a crucial stage in identity development, serving as a period for exploring one's future roles, values, and social position (Erikson 1988). During this period, individuals experiment with multiple possibilities and re-examine existing beliefs and norms. Erikson (1988) explained that adolescents form their identity through evaluating values and norms conveyed by social environments such as family, school, and media. Additionally, Blos (1967) described adolescence as the “second individuation stage,” emphasizing the emergence of an independent identity distinct from family. Because identity formation occurs through continuous interaction with social and environmental contexts, technological environments can also serve as important spaces where identity is constructed and negotiated. Digital technologies provide adolescents with expanded opportunities to access information, engage in social interaction, and experiment with different forms of self-expression (Katz and Aakhus 2002). At the same time, these environments may also introduce challenges such as information overload, fragmented self-presentation, and identity confusion (Israelashvili et al. 2012; Misra and Stokols 2012).

In identity formation theory, creative activity plays a particularly important role, as it allows individuals to express their thoughts, values, and abilities while constructing a sense of self through meaningful production (Hamman and Hendricks 2005). For adolescents, engaging in creative activities provides opportunities to explore their capabilities, communicate socially, and negotiate their identities within broader cultural contexts. In this context, AI differs from conventional technologies in its capacity to more actively intervene in the identity formation process through interaction and co-generation. Specifically, generative AI can influence how individuals perceive their roles and contributions by directly participating in content creation and decision-making processes (Schreiber, 2024).

However, in AI-based creative environments, the structure of creative processes is fundamentally transformed. AI can intervene in multiple stages of creation, including idea generation, image production, and writing, thereby blurring the boundaries of authorship and contribution (Schreiber, 2024). As a result, learners must continuously interpret and renegotiate their roles in relation to AI during the creative process. Through these interactions, learners may reconstruct their identities not merely as technology users, but as creators. By comparing and interpreting AI-generated

outputs alongside their own contributions, learners engage with the question of “who is the creator”, leading to a redefinition of their roles. This process reflects a core mechanism of identity formation, as identity is shaped through ongoing interaction with social and technological contexts (Erikson 1988; Katz and Aakhus 2002).

Importantly, learners' perceptions of AI's role influence this process in different ways. When AI is perceived as dominant in the creative process, learners may experience a diminished sense of control and perceive their creator identity as threatened (Kang and Lou 2022; Jiang et al. 2024). In contrast, when AI is understood as a supportive tool, learners are more likely to maintain agency and strengthen their sense of identity through perspectives of augmentation and human–AI collaboration (Raisch and Krakowski 2021).

Despite these emerging insights, research examining how learners' creator identity develops and transforms in AI-based creative environments, and how these identity processes relate to their perceptions of and reactions to AI remains limited.

2.4 Ethical and authorship challenges in AI-generated content

As AI technology advances, related research is being actively discussed in the field of education. However, existing AI-related research has often focused on effectiveness, performance, learning outcomes, and practical applications of AI technologies (Chiu et al. 2023). While these studies have contributed to demonstrating the potential for educational applications of AI, the ethical, social, and identity issues have received relatively less attention. In particular, the emergence of generative AI raises the need for new discussions, as AI goes beyond being a mere tool and directly participates in content creation and (Schreiber, 2024; Kariyawasam 2020).

In this context, existing research on AI ethics has largely been discussed from a principle-based perspective, focusing on fairness, transparency, accountability, and explainability (Floridi 2019; Hagendorff 2022; Munn 2023; Wiese et al. 2025). These discussions are grounded in concerns that AI may cause broader societal issues such as labor displacement, inequality, and the spread of misinformation (Nguyen and Hekman 2024). However, these discussions have primarily focused on norms and principles at the system level, while the ways in which such ethical issues are experienced and interpreted by students, particularly in creative contexts, remain less examined.

Generative AI differs from previous technologies in which it intervenes in domains traditionally considered uniquely human, such as creativity and authorship. These characteristics lead to the perception of AI not merely as a technical tool, but as an active role in the creative process,

which can create tensions (Schreiber, 2024). Consequently, AI-based creation shifts the focus from simply using technology to examining how creation is defined and understood. One of the central questions in AI-generated content is the issue of authorship, particularly the question of “who is the creator” (Hristov 2016; Huang et al. 2022). Although AI can make substantial contributions to the creative process, it is generally not recognized as an author under current academic and legal standards (Lund and Naheem 2024). As a result, the boundaries of contribution and responsibility among humans, AI systems, and developers become increasingly blurred (Hajibayova and Lee 2025). In addition, while discussions are underway regarding declaration policies specifying the use of AI and authorship attribution standards, consistent standards have not yet been sufficiently established in the contexts of creative writing and education (Hajibayova and Lee 2025; Lund and Naheem 2024). In addition, AI-generated content presents significant challenges to existing copyright systems. Issues such as ownership of AI-generated outputs, the use of training data, and the difficulty of determining originality remain unresolved. Given that AI-generated content is created based on existing data, the debate over whether such results can be regarded as genuine creation continues (Hajibayova and Lee 2025).

These ethical considerations are not limited to general issues but are closely related to learners’ experiences in AI-based art education. When learners perceive AI as replacing human creativity, they may experience concerns about a loss of control (Jiang et al. 2024; Kang and Lou 2022). Therefore, ethical issues in AI-based art education need to be understood, as they are closely connected to technology acceptance, agency, and identity formation, particularly in how learners interpret their roles and contributions (Jiang et al. 2024; Kang and Lou 2022). However, research on learners’ ethical tensions and perceptions of authorship in AI-based creative environments remains limited, particularly in terms of how these experiences connect to attitudes and reactions toward AI and students’ future career.

2.5 Research gap

Rapid advancements in AI are transforming creative work across disciplines, influencing not only how creative products are generated but also how creators understand their roles within the process. While AI-integrated tools increasingly mediate artistic production, far less is known about how sustained engagement with AI influences students’ evolving sense of themselves as creators.

Existing research has largely focused on the functional affordances of AI in creative contexts, such as increased efficiency, productivity, and idea generation. Studies frequently examine technical skill acquisition, workflow optimization, or learning outcomes associated with AI-supported

environments (Chiu et al. 2023). However, these instrumental perspectives overlook a deeper and more complex dimension: how collaboration with AI negotiates students’ creative identity, perceived autonomy, and authorship.

From the perspective of identity formation theory (Erikson 1980; 1988), identity is not a fixed trait, but a dynamic construct shaped through social interaction and environmental engagement. As students interact with AI systems that generate, suggest, and modify creative outputs, they may renegotiate their self-concept as creators. AI may be positioned as a neutral tool, an active collaborator, or even a competing agent that challenges human originality. These shifting positions can influence students’ sense of agency, creative ownership, and long-term career outlook.

Furthermore, while prior research has examined self-efficacy in technology-enhanced learning, limited attention has been paid to how AI-supported creative experiences simultaneously affect technical confidence, creative autonomy, and broader career perceptions. The tension between increasing AI competence and emerging concerns about job displacement, skill erosion, or copyright and authorship raises important questions about how students reconcile empowerment with uncertainty.

In addition, ethical concerns surrounding AI-generated art, including questions of originality, intellectual property, and the encroachment of machine systems into uniquely human domains, remain underexplored in relation to students’ identity negotiation processes. How students interpret these ethical tensions may shape not only their attitudes toward AI but also their willingness to embrace or resist AI-integrated creative practices.

To address these gaps, the present study examines how students conceptualize AI within their creative processes and how these perceptions relate to changes in self-efficacy, creative autonomy, and career outlook. By integrating quantitative measures of AI-related competence and perception with qualitative analyses of students’ reflections and classroom interactions, this study offers a multidimensional account of how AI participation intersects with creator identity formation in educational settings.

3 Methods

3.1 Participants and contexts

Participating students in this study were 84 tenth-grade students enrolled in a public high school located in a metropolitan city in South Korea. The participants included 21 male and 63 female students, aged 15 to 16 years. Most students reported little or no prior experience with AI-generated images ($n = 75$, 89.3%). The participating school is a specialized arts-focused high school with an emphasis on

animation, media creation, and related fields. The school emphasizes nurturing creative individuals who integrate imagination, artistic sensibility, and intellectual growth. Accordingly, many students at the school demonstrate strong interests in creative production and artistic expression.

The classroom implementation was led by one high school teacher with 23 years of teaching experience. The teacher has extensive expertise in educational technology and has actively integrated digital tools into her instructional practice. She is also the author of a practice-oriented book on AI integration for teachers, reflecting her professional experience and leadership in technology-enhanced teaching.

Institutional Review Board approval was obtained from the authors' institution. All participants provided informed consent for participation and academic publication, including authorization to use survey responses, interview data, and photographs of classroom activities.

3.2 Classroom curriculum

The curriculum was developed using a design-based research approach, characterized by iterative design, implementation, and refinement in collaboration with classroom teachers and the research team (Design-Based Research Collective 2003) using the Lee and Yu (2026) framework. This approach enabled the development of a learning environment that was both theoretically grounded and responsive to the realities of classroom practice. The intervention was implemented over the course of one academic semester and consisted of 10 weekly instructional sessions. Across the semester, the curriculum was intentionally structured to progressively support students' development of AI understanding, creative application, and critical reflection. To support implementation consistency, all sessions followed a common instructional sequence and used the same core learning activities, AI tools, and project structure within the same course context.

In the early phase of the intervention, the first three sessions focused on building foundational knowledge of AI. During this period, students were introduced to core concepts underlying AI systems, including how generative AI produces outputs and processes information. Following

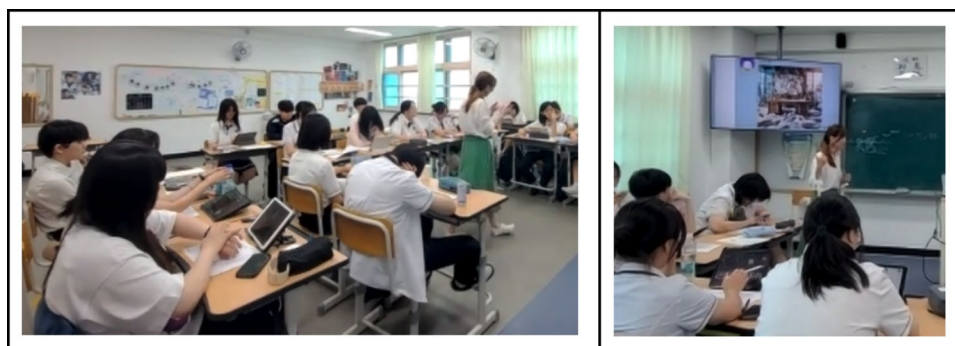
this, session four shifted toward future-oriented thinking, encouraging students to explore the potential impact of AI on society and creative industries. These sessions also invited students to reflect on possible future career paths in relation to emerging technologies. Next, spanning sessions five through seven, centered on hands-on creative practice. Students engaged with generative AI tools such as Canva and Adobe Express to produce visual artifacts, with particular emphasis placed on experimentation, iteration, and the integration of AI into their creative workflows. In the later phase, sessions eight and nine were dedicated to student presentations and peer feedback. Students shared their AI-assisted creative works, articulated their creative processes, and reflected on the role of AI within their projects. The final session served as a wrap-up, providing structured opportunities for guided reflection on their overall learning experiences, including their evolving perceptions of AI, their identities as creators, and related ethical considerations. Overall, the curriculum was designed not only to enhance students' technical competence in using AI tools but also to foster a deeper critical understanding of AI's role in creative work, support the development of their emerging creator identities, and cultivate ethical awareness in AI-mediated contexts (Figs. 1, 2).

3.3 Research design and data source

This study employed a mixed-methods design to examine how students' perceptions, identity, and ethical reasoning related to AI evolved through participation in an AI-integrated art course (Creswell and Creswell 2017). A convergent design was adopted, in which quantitative and qualitative data were collected within the same instructional context and integrated during interpretation to provide a comprehensive understanding of students' experiences. Quantitative findings were used to identify broad patterns in students' AI-related perceptions, while qualitative findings were used to contextualize and explain those patterns.

Quantitative data were collected through pre- and post-surveys. The survey measured four constructs: students' self-efficacy in AI knowledge and skills, perceptions of AI's

Fig. 1 Classroom activity



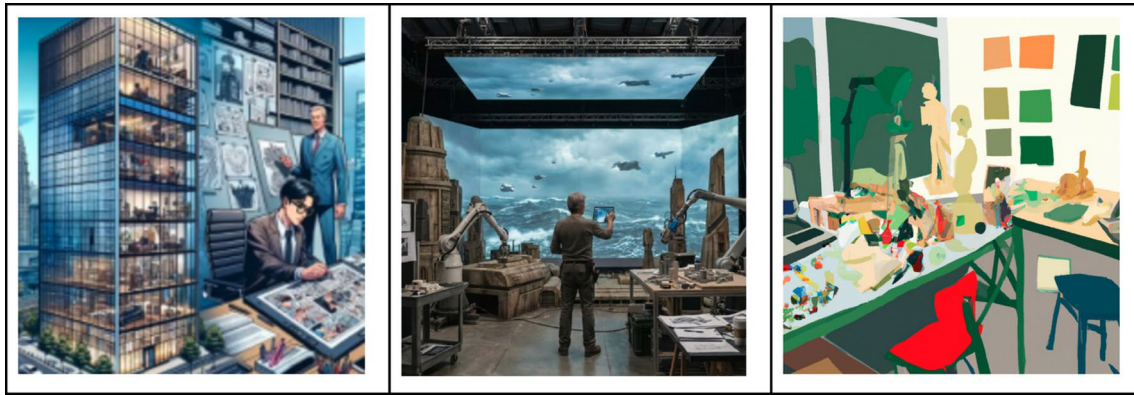


Fig. 2 Student artwork using generative AI

impact on work, ethical beliefs regarding AI-assisted art, and career-related attitudes. The instrument was adapted from prior studies, including Chiu et al. (2025) and Wang and Chuang (2024) for AI self-efficacy, Park et al. (2024) for perceptions of AI's impact on work, Lee et al. (2025) and Mok et al. (2025) for ethical beliefs regarding AI-assisted art, and Fu (2026) and Testa et al. (2026) for career-related attitudes. Items were revised for high school students and the art education context and then reviewed by experts in educational technology and art education. Cronbach's alpha coefficients for the four subscales demonstrated acceptable to high internal consistency, ranging from 0.78 to 0.91 in the pre-survey and from 0.80 to 0.93 in the post-survey.

To complement the survey data and capture the complexity of students' experiences, multiple sources of qualitative data were collected. Classroom observations were conducted throughout the intervention and documented through field notes focused on students' interactions with AI tools, participation patterns, creative decision-making, and emerging ethical concerns. In the final session, all students completed open-ended written reflection prompts about their experiences with AI-assisted artmaking, their perceptions of AI's role in creative work, their identities as creators, and related ethical considerations. In addition, semi-structured focus group interviews were conducted with 28 voluntary participants across five classes, with five to seven students participating from each class. Each focus group lasted approximately 30 min and was conducted after the final reflection session to further explore students' experiences, perceptions, and reasoning. Student artifacts, including AI-assisted creative outputs were collected and reviewed as supplementary contextual evidence. These artifacts helped contextualize how students integrated AI tools into their creative processes. These multiple quantitative and qualitative data sources enabled triangulation and provided a richer understanding of how students experienced and interpreted AI in creative contexts.

3.4 Data analysis and trustworthiness

Quantitative data from the pre- and post-surveys were analyzed using paired-samples t-tests to examine changes in students' AI-related perceptions and self-efficacy. Before conducting the paired-samples t-tests, the normality of the difference scores was examined, and no substantial violations were identified. Qualitative data were analyzed using thematic analysis (Braun and Clarke 2006). The analysis followed a hybrid inductive-deductive approach. Initial coding was guided by the study's conceptual focus on creator identity, agency, and ethical awareness, while additional codes were developed inductively from students' written reflections, focus group interviews, and observational notes. Three researchers participated in the coding process. The researchers first reviewed the qualitative data, generated initial codes, and then discussed coding differences through analytic meetings until consensus was reached. Themes were refined by comparing codes across data sources, revisiting representative excerpts, and ensuring coherence within themes and clear distinctions across themes.

Particular attention was given to how students described AI as a tool, collaborator, or source of uncertainty, and how these conceptualizations related to their perceptions of creator identity, agency, and ethical awareness. To enhance trustworthiness, analytic memos and an audit trail were maintained throughout the coding process to document initial codes, code revisions, theme development, and key analytic decisions. Because the research team was involved in the design and implementation of the AI-integrated art curriculum, reflexivity was addressed by documenting researcher assumptions and checking emerging interpretations across interview, reflection, observation, and artifact data.

4 Results

4.1 Changes in AI perceptions and self-efficacy

To examine changes in students' AI-related perceptions and self-efficacy, paired-samples t-tests were conducted comparing pre- and post-survey responses. Before conducting the paired-samples t-tests, the normality of the difference scores was examined. No substantial violations were identified. Table 1 shows the results across four domains.

Results indicate statistically significant improvements in three of the four AI self-efficacy items. Students reported increased ability to explain how AI generates outputs (Item 2; $t = -2.345$, $p = 0.023$), understand how AI recognizes images and language (Item 3; $t = -2.312$, $p = 0.025$, $d = 0.25$), and effectively use AI-based programs or applications (Item 4; $t = -2.542$, $p = 0.014$, $d = 0.28$). Although students' confidence in using AI to obtain desired results (Item 1) increased from pre- to post-test, the change was not statistically significant ($t = -1.659$, $p = 0.103$). Overall, these findings suggest that participation in the AI-integrated learning experience was associated with strengthened procedural and conceptual understanding of AI technologies.

Students' beliefs about AI's potential to replace human jobs (Item 5) slightly decreased from pre- to post-test, though this change was not statistically significant ($t = 1.815$, $p = 0.075$). Similarly, perceptions that AI would significantly impact human occupations (Item 6) remained largely unchanged ($t = -0.198$, $p = 0.844$). These results suggest that while students developed greater

AI competence, their broader beliefs about AI's societal and occupational impact remained relatively stable.

No statistically significant changes were observed in students' ethical perceptions of AI-assisted creative work (Items 7 & 8). Both the pre- and post-surveys indicate that students consistently viewed art as an inherently human domain. Agreement with the view that uniquely human domains such as art should not be encroached upon by AI remained stable. These findings underscore the persistence of students' deeply held belief that artistic creation should remain uniquely human.

Students' overall career optimism (Item 9) did not change significantly ($t = 0.248$, $p = 0.805$). However, persistence in the face of career challenges (Item 10) showed a statistically significant decrease ($t = 2.388$, $p = 0.020$, $d = 0.26$), indicating a slight reduction in self-reported perseverance following the intervention. The decrease in Item 10 shows that, as students developed a more concrete understanding of AI's potential impact, they may have engaged in more critical reflection about their future careers.

These findings indicate that while the AI-integrated learning experience enhanced students' technical competence, it did not lead to corresponding changes in their broader beliefs or attitudes. Instead, the results point to emerging tensions related to creator identity, ethical judgment, and future career expectations. To further examine these tensions, the following sections draw on qualitative data to explore how students make sense of the role of AI in creative processes (ontology), their identity as creators (agency), and the ethical implications of AI-assisted creation (ethics).

Table 1 Survey results

Component	Question	Pre	Post	t	p
Self-efficacy in AI knowledge and skills	1. I can use AI to obtain the results I need	3.455	3.673	-1.659	0.103
	2. I can explain how AI generates its outputs	3.036	3.382	-2.345	0.023
	3. I can understand how AI recognizes images and language	3.073	3.455	-2.312	0.025
	4. I can effectively use programs or applications that incorporate AI technologies	3.091	3.473	-2.542	0.014
Perceptions of AI impact on work	5. I believe that in the future, a large portion of human jobs will be replaced by AI	4.200	3.945	1.815	0.075
	6. believe that AI will have a significant impact on human occupations	4.473	4.491	-0.198	0.844
AI-assisted art and ethics	7. I believe that artworks created with the help of AI can be recognized as a form of art	2.036	2.236	-1.334	0.188
	8. I believe that art is a unique human domain that cannot be replaced by machines and AI technology	4.127	4.145	-0.134	0.894
Career pathway	9. I feel positive about my future career	3.909	3.873	0.248	0.805
	10. Even when I face challenges in my career path, I will persist in art	3.945	3.636	2.388	0.020

4.2 Conceptual typologies of AI perception in creative work

To address the second research question, this study examined how students conceptualize the role of AI in the creative process. Analysis of student responses revealed three dominant orientations: AI as a tool, a collaborative partner, and a potential threat (Fig. 3). These categories are not mutually exclusive but rather represent a continuum of perception through which students interpret AI's role in creative work.

4.2.1 Tool-oriented perspectives: AI as a productivity tool

A substantial number of students conceptualized AI primarily as a functional tool that enhances efficiency and reduces the burden of repetitive or time-consuming tasks. In this view, AI is positioned as an assistive mechanism that supports, rather than alters, the creative process. Students frequently emphasized speed, convenience, and the ability to externalize mental imagery:

“With just a few words, it quickly and conveniently created the drawings I wanted.” (Student ID 1108)
 “It helped visualize the images I was imagining quickly.” (Student ID 1219)

Others highlighted AI's role in offloading tedious preparatory work:

“It handles tasks like sketching outlines or basic groundwork that I usually find tedious, which saves time and makes the process more efficient.” (Student ID 1502)

“AI helps with tasks that are close to manual labor, making the process easier.” (Student ID 1521)

In addition, some students associated AI with improved output quality:

“AI can be a tool that produces higher-quality results.” (Student ID 1401)

These responses position AI as an instrumental extension of human capability, where efficiency gains are foregrounded while authorship remains firmly human-centered.

4.2.2 Partner-oriented perspectives: AI as a creative partner

Beyond a purely instrumental role, many students described AI as an active collaborator that contributes to idea generation and creative exploration. In this orientation, AI is not merely executing tasks but participating in the creative process by offering suggestions, feedback, and unexpected outputs. Students described using AI as a source of inspiration:

“Sometimes, when I struggle with inspiration, I enter keywords into AI and give specific commands. It helps me brainstorm ideas for stories or illustrations.” (Student ID 1425)

“There were moments when I got unexpectedly good ideas from AI.” (Student ID 1529)

Others framed AI as a supportive collaborator:

“AI can act as a supporter that provides feedback and ideas.” (Student ID 1129)

“Humans and AI can collaborate to create better work.” (Student ID 1318)

In this perspective, creativity is understood as a distributed process, where human intention and machine-generated suggestions interact dynamically. AI is thus positioned as a co-creative agent that expands, rather than replaces, human ideation.

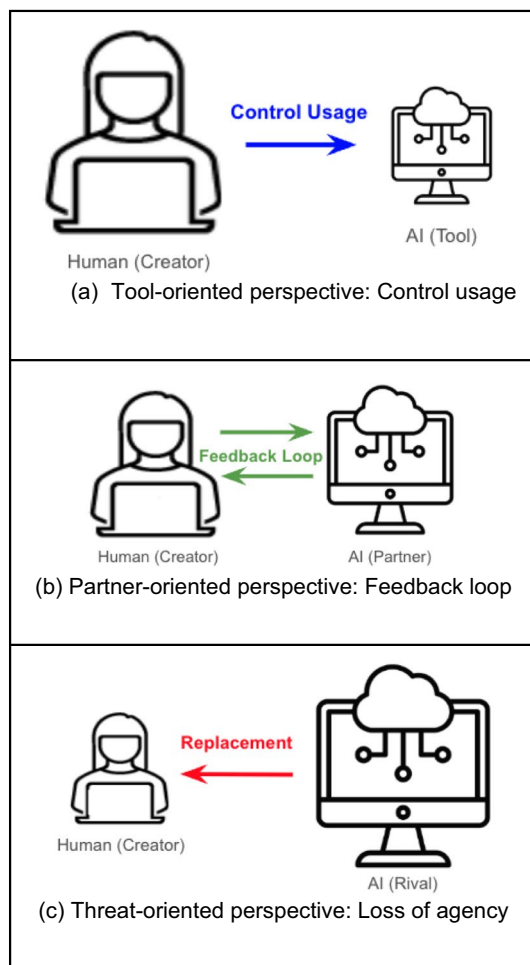


Fig. 3 Conceptual typologies of AI perception in creative work

4.2.3 Threat-oriented perspectives: AI as a threat to creativity and human labor

In contrast, a third group of students expressed concern, discomfort, or even strong resistance toward AI in creative contexts. These perceptions framed AI as a disruptive force that may undermine creativity, human agency, and the value of artistic labor. Some students described an initial sense of awe accompanied by unease:

“It was fascinating to use AI for the first time, but also a bit scary to see how advanced it already is.” (Student ID 1306)

Others articulated concerns about the pace of technological advancement:

“The speed at which AI is developing is almost frightening, and it seems difficult for individuals to keep up.” (Student ID 1231)

A recurring theme was the fear that efficiency may come at the expense of artistic depth:

“I’m worried that in creative fields, efficiency will increase while artistic value decreases.” (Student ID 1432)

Students also expressed concerns about dependency and skill erosion:

“If I use AI too much, I might forget how to come up with ideas on my own.” (Student ID 1310)

Finally, several students raised concerns about labor displacement and the devaluation of human artists:

“If AI can create professional-quality illustrations in seconds, why would anyone pay an artist?” (Student ID 1409)

“If AI keeps improving, will there still be a need for human artists?” (Student ID 1406)

These responses reflect broader anxieties about automation, suggesting that students perceive AI not only as a tool or partner but also as a potential competitor that challenges the sustainability of human-centered creative work.

4.3 Creator identity and perceived autonomy

To address the third research question, this study examined how AI integration negotiates students’ perceptions of creator identity and agency. Analysis revealed a continuum of perceived autonomy, ranging from strong human-centered control to diminished agency under AI influence (Fig. 4).

4.3.1 High autonomy: AI as assistive support

Students in this group maintained a strong sense of authorship and viewed AI as a supplementary tool that enhances efficiency without compromising creative control.

“I still make all the creative choices. AI just helps with small things, like generating references or basic coloring.” (Student ID 1547)

Some students explicitly rejected the idea that AI poses a significant threat:

“After using AI, I realized it’s not as powerful as I expected. The results weren’t always satisfying, and it didn’t feel like it could replace artists. It’s useful for simple tasks, but not a real substitute.” (Student ID 1231)

For these students, creativity remains fundamentally human-driven, with AI serving as a limited augmentation tool.

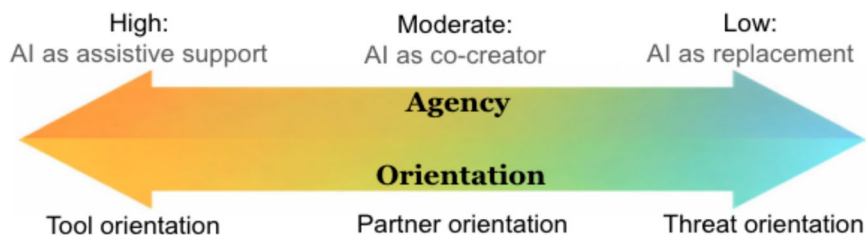
4.3.2 Moderate autonomy: AI as co-creator

Students in the moderate autonomy group described a more entangled relationship with AI, where creative control is shared, negotiated, and sometimes dependent on AI-generated suggestions.

“AI helped structure my ideas, but I still edit what it gives me, and the final decision is mine.” (Student ID 1309)

Here, authorship becomes collaborative. While students retain ultimate control, they acknowledge that AI meaningfully shapes the direction and development of their work.

Fig. 4 Agency and orientation spectrum



4.3.3 Low autonomy: AI as replacement

Students with low perceived autonomy expressed concern that AI may displace their role as creators or diminish their creative agency.

“I used to draw everything from my imagination. Now I start with AI, and it creates things for me. I’m not sure if that makes me a better artist or a lazier one.” (Student ID 1107)

“I feel anxious that rapid advancements in AI may prevent me from securing a job in the future.” (Student ID 1407)

This perspective reflects anxiety about over-reliance on AI and the potential erosion of foundational creative skills. For these students, AI introduces tension into their identity as creators, raising questions about authenticity, effort, and ownership.

4.4 Ethical tensions and identity negotiation

The fourth research question examined how ethical concerns surrounding AI intersect with students’ acceptance of AI and their self-perception as future creators. Findings indicate that ethical considerations are deeply intertwined with identity formation and significantly shape students’ attitudes toward AI.

A central concern involved authorship and originality. Students questioned whether AI-generated outputs could be legitimately attributed to themselves:

“AI-generated art looks too similar to real artists’ work, which makes me uncomfortable and feels unethical.” (Student ID 1526)

“What AI creates cannot be considered art. It feels like mass-produced results rather than genuine creation.” (Student ID 1104)

Transparency in AI training data emerged as another major issue:

“We need transparent processes, especially about how AI learns from data and existing artwork.” (Student ID 1422)

“AI is trained on other people’s work. When I use it, am I unknowingly stealing?” (Student ID 1513)

Some students articulated strong ethical rejection of AI in creative contexts:

“Training AI on copyrighted artwork without consent is not acceptable. Machines cannot engage in artistic practice.” (Student ID 1327)

“Using other artists’ work through AI to generate outputs is not art. It feels unethical.” (Student ID 1235)

Others framed AI use as a form of unfair advantage:

“In situations where people should be evaluated based on their own abilities, using AI is like cheating.” (Student ID 1134)

Beyond legal concerns, students also questioned whether AI can possess the essential qualities of creativity:

“AI doesn’t have feelings, memories, or experiences. How can it create real art?” (Student ID 1110)

“Please don’t describe AI as drawing.” (Student ID 1528)

“The moment we give AI the status of a creator, the boundaries of art collapse.” (Student ID 1535)

These responses suggest that some students were hesitant to recognize AI as a legitimate creative agent. Ethical concerns, therefore, are not merely external considerations but are central to how students define creativity, authorship, and their own identities as creators.

5 Discussion

This study examined how AI-integrated creative learning environments influence students’ self-efficacy, perceptions of AI, and the formation of creator identity. By integrating quantitative and qualitative findings, the results reveal a complex pattern: while students developed stronger AI-related competencies, their perceptions of creativity, authorship, and future careers became more nuanced.

5.1 AI competence and acceptance

The quantitative findings demonstrate clear gains in AI-related self-efficacy, particularly in students’ understanding of how AI systems function and their ability to use AI tools effectively. These results suggest that the instructional intervention successfully supported both conceptual and procedural engagement with AI (Lee and Kwon 2024).

However, although students increased their AI knowledge, these gains did not translate into more optimistic perceptions of AI’s societal impact or positive attitudes toward AI-generated artwork. This implies that AI competence does not necessarily correspond to AI acceptance, since they may be influenced by various contextual factors (Jiang et al. 2024; Shin 2026a). Students’ beliefs about AI’s influence on jobs remained largely unchanged, and notably, career persistence decreased significantly. This divergence suggests that increased competence does not lead to increased confidence

in the future. Instead, greater exposure to AI may prompt more critical and realistic reflections on career stability and competition.

The survey findings suggest that AI education increased students' technical competence without leading to uncritical acceptance of AI in creative work. While students demonstrated significant gains in self-efficacy in AI skills, their broader perceptions of AI remained differentiated and, in some cases, resistant. Notably, students' agreement with statements suggesting that AI would fully replace human jobs decreased after hands-on experience, whereas their belief that AI would have a substantial impact on occupations remained consistently high. These patterns indicate that students came to distinguish between technological replacement and technological influence, recognizing AI as unlikely to fully substitute human creators while still capable of reshaping work processes, particularly in terms of efficiency and task distribution.

5.2 From AI as Tool to AI as ontological question

Qualitative findings further reveal that students do not hold a single, unified perception of AI. Instead, they position AI along a continuum as a tool, a collaborator, or a threat. These varying conceptualizations reflect not only functional interpretations but also deeper ontological stances regarding the nature of creativity and authorship.

Importantly, these positions are not merely descriptive but shape how students engage with AI in practice. Viewing AI as a tool maintains a clear boundary between human and machine, whereas seeing AI as a partner introduces a more distributed model of creativity. In contrast, perceiving AI as a threat reflects a breakdown of this boundary, where AI is seen as encroaching upon or replacing human creative roles.

5.3 Negotiation of creator identity and career persistence

These differing perceptions of AI roles are closely tied to variations in perceived agency and autonomy. Students who viewed AI as a tool maintained strong agency and a stable sense of authorship, while those who perceived AI as a collaborator experienced shared or negotiated control. In contrast, students who viewed AI as a threat reported diminished autonomy and expressed concern about losing their role as creators.

This continuum suggests that AI integration did not appear to uniformly enhance or diminish students' sense of agency; rather, it introduced new conditions under which students perceived and negotiated their agency in AI-assisted creative work. Students are not simply adopting AI tools but actively renegotiating what it means to be a creator in an AI-mediated environment. In this sense, AI appeared to

serve as a context through which students reflected on and negotiated their roles and identity as creators in AI-assisted creative processes (Erikson 1988; Katz and Aakhus 2002), prompting reflection on authorship (Hristov 2016; Huang et al. 2022), originality, and the value of human contribution (Hajibayova and Lee 2025). Identity formation theory suggests that adolescents' sense of self develops through interaction with social and technological contexts (Erikson 1988; Katz and Aakhus 2002). For aspiring artists, working with AI raises fundamental questions such as "*Am I still the creator?*" and "*What does authorship mean in AI-assisted art?*".

Interestingly, students reported a decrease in unconditional career persistence in the survey, indicating a shift away from idealized narratives of perseverance. Rather than signaling diminished motivation, this change may reflect increased realism about the evolving structure of creative work in an AI-influenced context. As students recognized that AI may not eliminate creative professions but could alter entry pathways and redistribute routine labor, they may have begun to reconsider a creative career. In this sense, AI education appears to have fostered reflective reconsideration rather than resignation, supporting a view of learning that emphasizes critical awareness over compliance. While students gain technical competence, they also encounter challenges to their assumptions about creativity, authorship, and future careers. The observed decrease in career persistence, therefore, should not be interpreted as a negative outcome or instructional failure. Rather, it may reflect a process of reflection, questioning, and negotiation regarding AI-assisted creativity and authorship, in which students move from simplified or optimistic assumptions toward more complex and critical understandings of AI's role in society. This tension may be a necessary condition for deeper learning. Instead of reinforcing uncritical optimism, effective AI education may require creating spaces where students can grapple with uncertainty, ambiguity, and ethical complexity.

5.4 Ethical tensions of identity formation

Ethical concerns emerged as a central dimension of this identity negotiation process. This concern aligns with broader debates in AI ethics and intellectual property authorship (Floridi 2019; Hajibayova and Lee 2025). Students' ethical and artistic boundaries remained firmly intact. Students had low acceptance for AI-created output as arts in both pre- and post-survey. This resistance aligns with the lack of change observed in survey items related to the irreplaceability of human creativity and the ethical legitimacy of AI-generated art. Following the interviews, students posed questions about authorship, copyright, and the legitimacy of AI-generated work, often linking these concerns directly to their own identity as creators.

Numerous students resisted attributing creative agency to AI, emphasizing the importance of human intention, experience, and emotional depth. These responses suggest that ethical considerations are not peripheral but foundational to how students define creativity itself. The rejection of AI as a legitimate creator reflects an attempt to preserve boundaries between human and machine in the face of increasingly blurred distinctions.

5.5 Implications for research in AI education

Our findings highlight the importance of distinguishing between competence and endorsement in AI education research. While students demonstrated significant gains in technical understanding and skills, these gains did not translate into uncritical acceptance of AI as a legitimate creative agent. Instead, students engaged with AI in a selective and critical manner, using it pragmatically while maintaining clear ethical and identity-based boundaries. This suggests that increased competence may, in some cases, heighten critical awareness rather than promote acceptance. These findings challenge implicit assumptions in AI education that skill development naturally leads to positive attitudes toward AI. Future research should therefore move beyond linear models of learning outcomes and examine the complex, and at times conflicting, relationships among competence, identity, and ethical judgment in AI-mediated learning environments.

5.6 Implications for art education in the age of AI

Our study highlights that AI-integrated instruction should move beyond a sole focus on technical skill development.

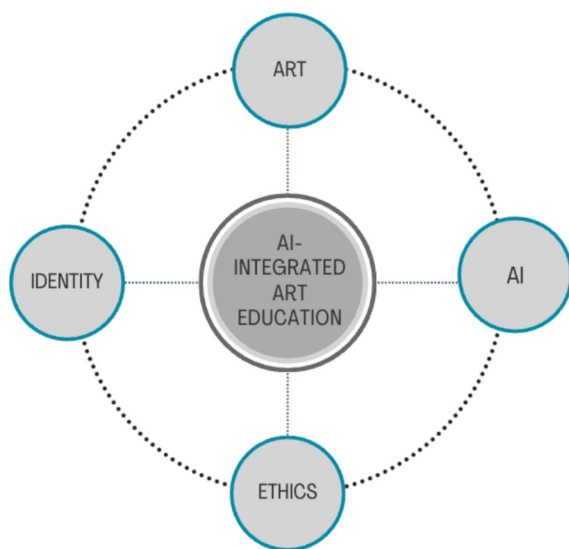


Fig. 5 Four elements in AI-integrated art education

Our findings suggest design principles and core components for AI-integrated art education (Fig. 5). In the context of art education, this involves a more comprehensive approach that addresses multiple dimensions of learning. First, foundational artistic skills and knowledge remain essential. AI should not replace core artistic competencies but rather build upon them. Second, students must develop active agency in working with AI. This includes not only learning how to use AI tools, but also how to direct, critique, and intentionally integrate AI into their creative processes. Third, art education should explicitly address creator identity. As AI becomes embedded in creative workflows, students need structured opportunities to reflect on authorship, ownership, and what it means to create in collaboration with non-human agents. Finally, ethical understanding must be treated as a core component rather than an add-on. Students' awareness of issues such as copyright and data use functions as a critical lens through which they evaluate the legitimacy of AI-generated work.

These findings call for a reorientation of art education toward balancing artistic skill, human agency, identity formation, and ethical reasoning. Without these components, students may develop technical competence without the conceptual and ethical frameworks needed to navigate AI-rich environments. These findings highlight the need for a more holistic model of AI-integrated art education that balances art skill development with identity formation and AI ethical understanding.

5.7 Implications for career education

This study contributes to emerging discussions on career sensemaking in the age of AI. Students did not perceive AI merely as a classroom tool, but as a force with the potential to influence creative labor and future career trajectories. Notably, increased exposure to AI was associated with reduced career persistence, suggesting that learning about AI may introduce uncertainty alongside adaptability. Students' responses reflected a dual process: growing awareness of AI's capabilities alongside concerns about their own future roles. These findings suggest that career education should move beyond promoting adaptability alone and instead support students in navigating ambiguity, uncertainty, and questions about their roles in rapidly evolving creative work environments.

6 Conclusion

This research reveals a deep ambivalence toward AI in creative work. Students recognize AI's potential for efficiency, inspiration, and collaboration, but they also struggle with issues of identity, career pathways, and ethics.

While some students see AI as an enhancement to their creative process, others feel it undermines their role as artists. These results suggest that AI literacy education must extend beyond technical proficiency. Schools must incorporate critical discussions on authorship, ethics, and career adaptability, helping students develop a balanced perspective on AI's role in creativity. Future research should explore how these attitudes evolve over time and what pedagogical strategies can best support students in integrating AI without compromising their creative identity. As AI continues to redefine artistic production, educators must ensure that young creators retain their agency, refine their artistic skills, and critically engage with the ethical challenges of AI-assisted creativity. Without such guidance, there is a risk that students will either over-rely on AI to the detriment of their own development or reject it entirely due to uncertainty and fear. A nuanced and well-structured approach to AI education will be essential in preparing students for a future where human and machine creativity coexist. The future research needs to implement classroom intervention with new curricula combined ethics and creator reformation in AI-assisted art education.

Author contributions H.L. conceptualized the study, conducted data collection, performed data analysis, and wrote the original draft. H.L. also contributed to revising the manuscript. W.Y. contributed to data analysis and original draft. J.L. contributed to conceptualization, data collection, data analysis, and original draft. All authors reviewed and approved the final manuscript.

Data availability The data of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that there are no competing interests.

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